Parity of a paired marking with two endpoint anchors

Human Compression project

We acquire the separately proved anchor-deletion theorem and the exact seven- and nine-point boundaries, including their distinct reviews. The infinite deduction below uses these actual action statements; no general near-half classification is taken as a premise.

Let $G = \langle h_0, \dots, h_{r-1} \rangle$ act transitively on $2r + 1$ points. The marks form a string C-marking. Every colour is paired: it has at least two edges with endpoints in different orbits of the corresponding maximal omission. A first anchor b is moved by h_0 and fixed by all other marks. A last anchor c is defined in the reversed order, and b, c are distinct.

The acquired anchor-deletion theorem says that for $r \geq 5$ the actual marks begin

$$h_0 = (b\ x)(u\ v), \qquad h_1 = (x\ u)(v\ z)\beta, \tag{1}$$

where β is supported in $V \setminus \{z\}$ and $V = \Omega \setminus \{b, x, u, v\}$. The actual tail $K = \langle h_2, \dots, h_{r-1} \rangle$ fixes the four removed points, is transitive on V, and retains the paired property, the intersection property and the two anchors z, c. Its degree and rank are $2(r-2)+1$ and $r-2$. For $r \geq 7$, the same theorem proves $\beta = 1$ by an actual subgroup-intersection argument. These conclusions do not assume that G or K is symmetric.

Theorem 1. *Every mark of G is even. In particular, no natural symmetric marking of degree $2r + 1$ and rank r can have the stated paired property and two endpoint anchors.*

Proof. We use induction on r. Rank one is impossible, since a single involution cannot be transitive on three points. At rank two each mark on five points has at least two transpositions, by eligibility, and at most two by the matching property. Thus both are even. At rank three the acquired seven-point anchor theorem excludes transitivity already from the string and anchor conditions.

The nine-point boundary supplies the complete rank-four result. Every transitive paired string marking with the two anchors is, after a point relabelling respecting the anchor order, the path whose successive edge labels are

$$0, 1, 0, 1, 2, 3, 2, 3. \tag{2}$$

In particular each mark is a double transposition and the generated group is A_9. Its intersection property and actual group are independently checked in that acquired boundary, although its classification of the transitive paired tuples does not require IP.

At rank five, (1) would give a rank-three seven-point tail with two anchors, which is impossible. Consider rank six. Its tail has rank four and degree nine, so the acquired literal path (2) applies. Use its point labels so the first tail mark is

$$h_2 = (z\ p)(q\ w).$$

The tail omission $D = \langle h_3, h_4, h_5 \rangle$ has orbits $\{z\}, \{p, q\}$ and a six-point orbit containing c. Its centralizer fixing z, c is exactly $\{1, (p\ q)\}$. The factor β in (1) belongs to this centralizer: it fixes z, c, and h_1 commutes with h_3, h_4, h_5, which fix x, u, v, z. Thus β is either 1 or $(p\ q)$.

In the latter case $(h_0 h_1)^5 = \beta$, since the other factor of this product is a five-cycle on b, x, u, v, z. Conjugation by h_2 gives $(z\ w)$. Therefore

$$\delta = (p\ q)(z\ w) \in \langle h_0, h_1, h_2 \rangle \cap K.$$

The second membership uses the actual $K = A_9$, since δ is even and supported in its nine-point orbit. The original IP makes this intersection $\langle h_2 \rangle$, whereas δ is neither its identity nor its nonidentity element. This contradiction gives $\beta = 1$. Both marks in (1) and every tail mark are now even.

For $r \geq 7$, the acquired theorem already gives $\beta = 1$. Induction on the actual tail shows that all its marks are even. The two remaining marks in (1) are double transpositions, proving the assertion. Both inductions decrease rank by two on the exact support; no projected subgroup is assigned an intersection property. □

Corollary 2. *Under the strict smaller-degree sufficient-split induction, the critical overlap of a six-point rank-three even A_5 factor and a fractured natural symmetric factor of even degree m and rank $m/2 + 1$ has an internal sufficient split.*

Proof. We use the exact configuration and definition of sufficient split in the acquired first-cut and endpoint manuscripts. The full degree is $n = m + 5$. The first anchored unique cut in the symmetric factor is perfect. If its index is positive, the complete first-cut theorem supplies a sufficient split, using only the stated strict smaller-degree induction. If its index is zero, the complete endpoint theorem either supplies such a split or produces an actual natural symmetric middle of degree $2k + 1$ and rank k, paired in every colour and with two distinct endpoint anchors. Theorem 1 excludes precisely that residual. Thus both routes finish. The rank induction of Theorem 1 is independent of the outer sufficient-split induction, so no circular premise is introduced. □

Attribution and scope. The source is Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. Its paired near-half classification includes (2). Here only the exact nine-point boundary is enumerated; the infinite parity deduction uses the separately proved anchor deletion and actual tail. The classical rank results and earlier support arguments remain attributed through their acquired proofs. This component excludes the indicated middle and closes a conditional sufficient-split route. The complete degree induction, the stable counting correspondence, contraction and reversal compatibility remain separate obligations. No whole-original, conjecture, Lean or compression credit is asserted.